

5G/6G Communication Systems- ECA5604 (LAB)

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Experiment 2: - Implementing TDMA Slot Allocation in GSM.

AIM: To implement TDMA (Time Division Multiple Access) slot allocation in GSM using MATLAB and analyse system performance in terms of time slot duration, frame utilization, and number of allocated slots.

Objective 1:

MATLAB Code:

```
clc; clear; close all;

FrameTime = 4.615; Slots = 8;

SlotDuration = FrameTime/Slots;

t = 0:SlotDuration:FrameTime;

figure

subplot(2,1,1)

stem(t,ones(size(t)),'filled')

title('TDMA Time Slot Structure')

xlabel('Time (ms)')

ylabel('Slot Level')

grid on

subplot(2,1,2)

plot(t,ones(size(t)),'-o','LineWidth',1.5)

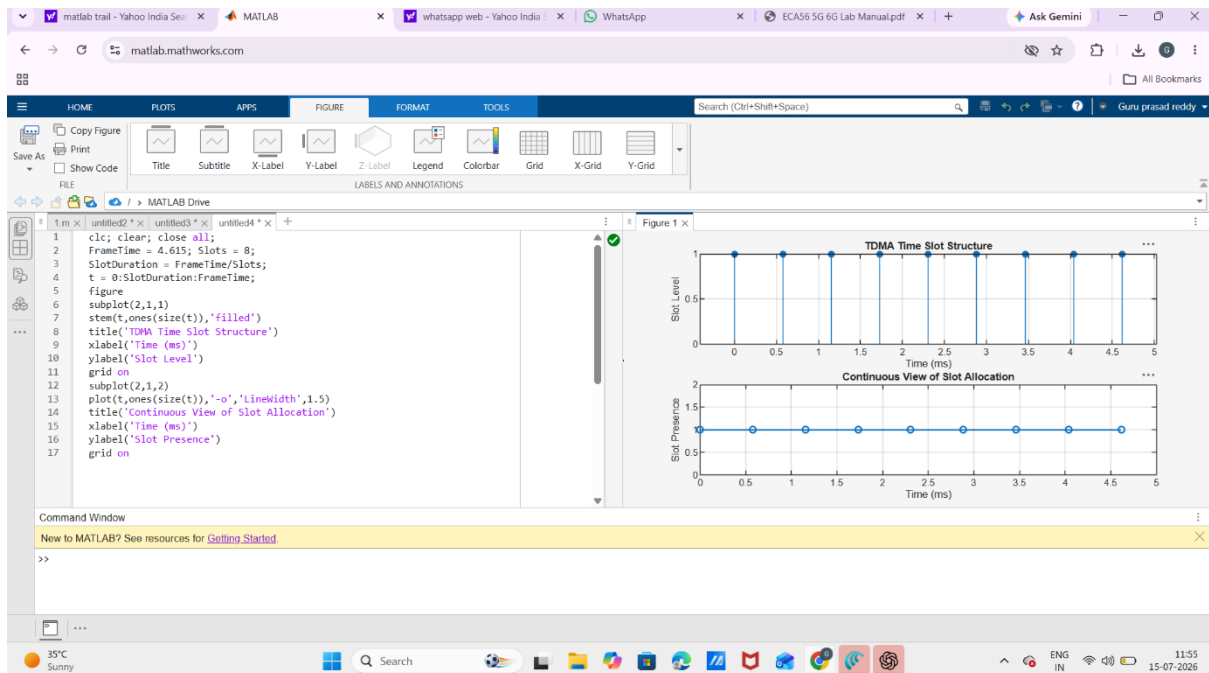
title('Continuous View of Slot Allocation')

xlabel('Time (ms)')

ylabel('Slot Presence')

grid on
```

Output:



Objective 2:

MATLAB Code:

```
clc; clear; close all;
```

```
TotalSlots = 8;
```

```
AllocatedSlots = 6;
```

```
FreeSlots = TotalSlots - AllocatedSlots;
```

```
Utilization = (AllocatedSlots/TotalSlots)*100;
```

```
disp('Frame Utilization (%)')
```

```
disp(Utilization)
```

```
figure
```

```
% Graph 1: Simple Line Plot (SAFE - no bar function)
```

```
subplot(2,1,1)
```

```
plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)
```

```
title('GSM Frame Utilization (Allocated vs Free)')
```

```
xlabel('Slot Type Index (1=Allocated, 2=Free)')
```

```
ylabel('Number of Slots')
```

```
grid on
```

% Graph 2: Pie Chart (SAFE)

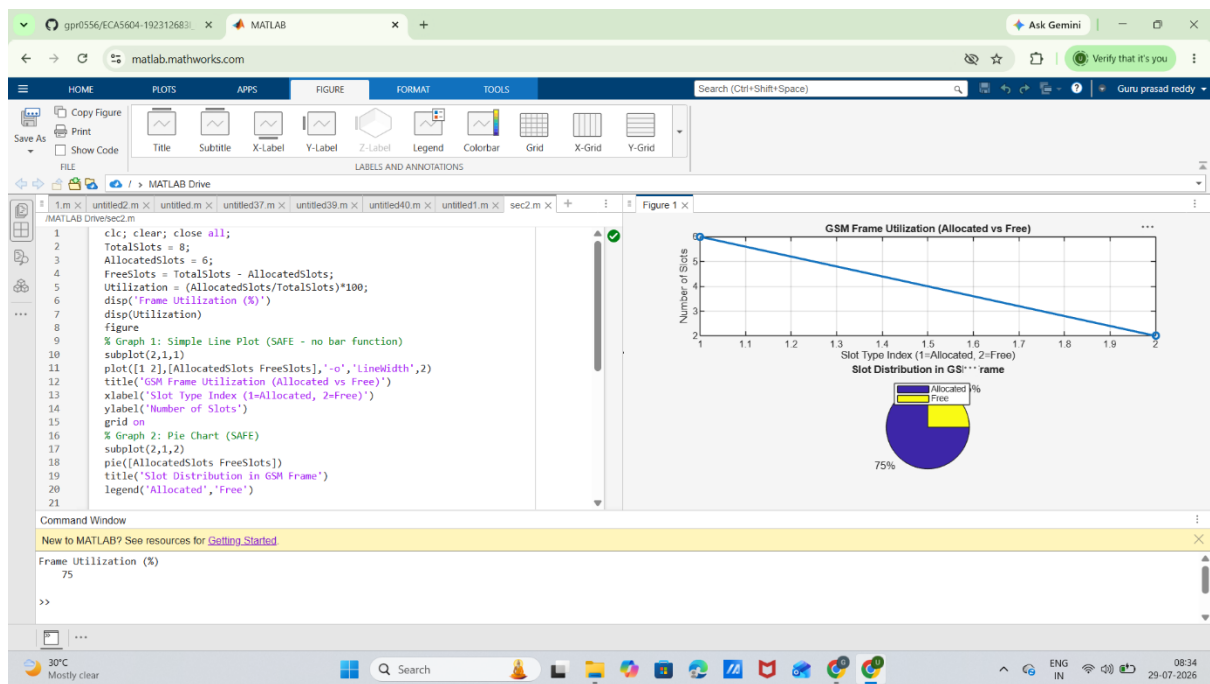
subplot(2,1,2)

pie([AllocatedSlots FreeSlots])

title('Slot Distribution in GSM Frame')

legend('Allocated','Free')

Output:



Objective 3:

MATLAB Code:

clc; clear; close all;

Users = 1:8; % Total users

TotalSlots = 5; % Available slots

Alloc = zeros(1,8); % Initialize allocation

Alloc(1:TotalSlots) = 1; % Assign slots to first users

Blocked = 1 - Alloc; % Remaining users blocked

disp('User Wise Slot Allocation')

disp(table(Users',Alloc','VariableNames',{'User','Status(1=Allocated,0=Blocked)'}))

figure

```
% Graph 1: Allocation per user

subplot(2,1,1)

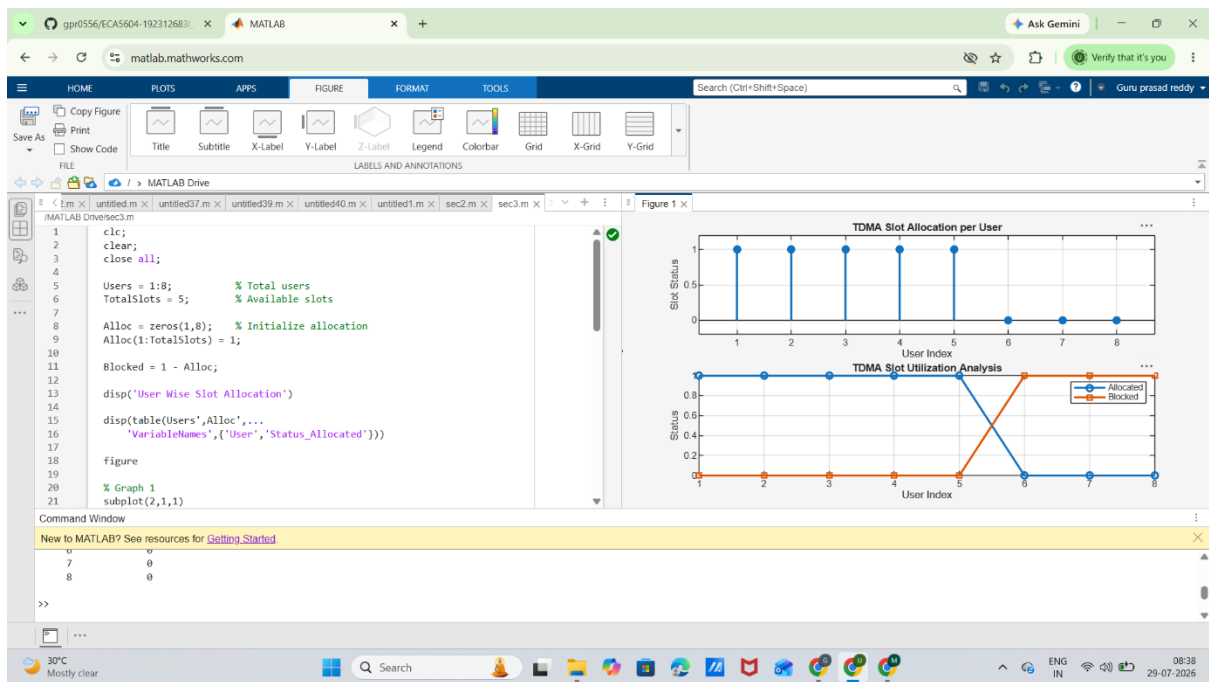
stem(Users,Alloc,'filled','LineWidth',2)
title('TDMA Slot Allocation per User')
xlabel('User Index')
ylabel('Slot Status (1=Allocated, 0=Blocked)')
ylim([-0.2 1.2])
grid on

% Graph 2: System view (safe alternative to bar)

subplot(2,1,2)

plot(Users,Alloc,'-o','LineWidth',2)
hold on
plot(Users,Blocked,'-s','LineWidth',2)
title('TDMA Slot Utilization Analysis')
xlabel('User Index')
ylabel('Status')
legend('Allocated','Blocked')
grid on
```

Output:



Result:

The TDMA slot allocation in a GSM system was successfully implemented using MATLAB. The time slot duration, frame utilization, and number of allocated time slots were calculated, demonstrating efficient and interference-free resource allocation among multiple users.